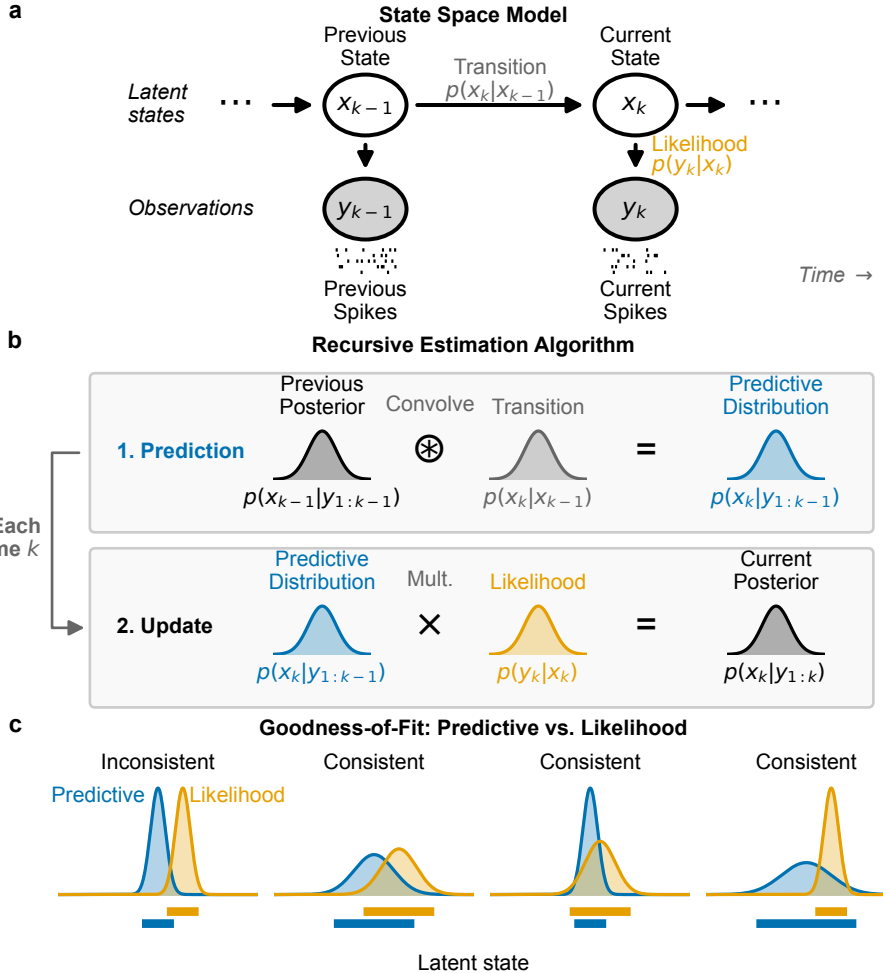

a
State Space Model
Previous State
Transition $p(x_k|x_{k-1})$
Current State
Latent states
x_{k-1}
x_k
Likelihood $p(y_k|x_k)$
Observations
y_{k-1}
y_k
Previous Spikes
Current Spikes
Time →
b
Recursive Estimation Algorithm
Each time k
1. Prediction
Previous Posterior
Convolve
Transition
Predictive Distribution
$p(x_{k-1}|y_{1:k-1})$
$p(x_k|x_{k-1})$
$p(x_k|y_{1:k-1})$
2. Update
Predictive Distribution
Mult.
Likelihood
Current Posterior
$p(x_k|y_{1:k-1})$
$p(y_k|x_k)$
$p(x_k|y_{1:k})$
c
Goodness-of-Fit: Predictive vs. Likelihood
Inconsistent
Consistent
Consistent
Consistent
Predictive
Likelihood
Latent state